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Ruhr-Universität Bochum
Experimental Physics I

WINTER 2026/27 COURSE 160202

Introduction to Nuclear and Particle Physics

Exercise sheet 4

Lecturer	Prof. Dr. Mikhail Mikhasenko
Assistant	Dr. Marian Stahl
Tutors	Hendric Jonas
	Dhruvanshu Parmar
Distributed	November 6, 2026
Due	November 13, in the lecture

Homework – problems from Thomson

Textbook. Mark Thomson, *Modern Particle Physics*, Cambridge University Press, first edition (2013).

Use the problem statements and figures in your copy of the textbook. Show your reasoning, state any assumptions, and include units in numerical answers.

Exercise 1 — Thomson 1.8

Radiation length X_0.

Exercise 2 — Thomson 1.9

Cherenkov radiation.

Exercise 3 — Thomson 2.7

The Λ baryon.

Exercise 4 — Thomson 3.1

Centre-of-mass frame energy.

Submission

Write your solutions by hand and work individually. Hand in your solutions on **November 13, 2026, at the lecture**. AI assistance is permitted for preparation; be ready to explain your reasoning independently.